4.18

(1) 一位全加器

设输入为 $A_0$、$B_0$、$C_{in}$

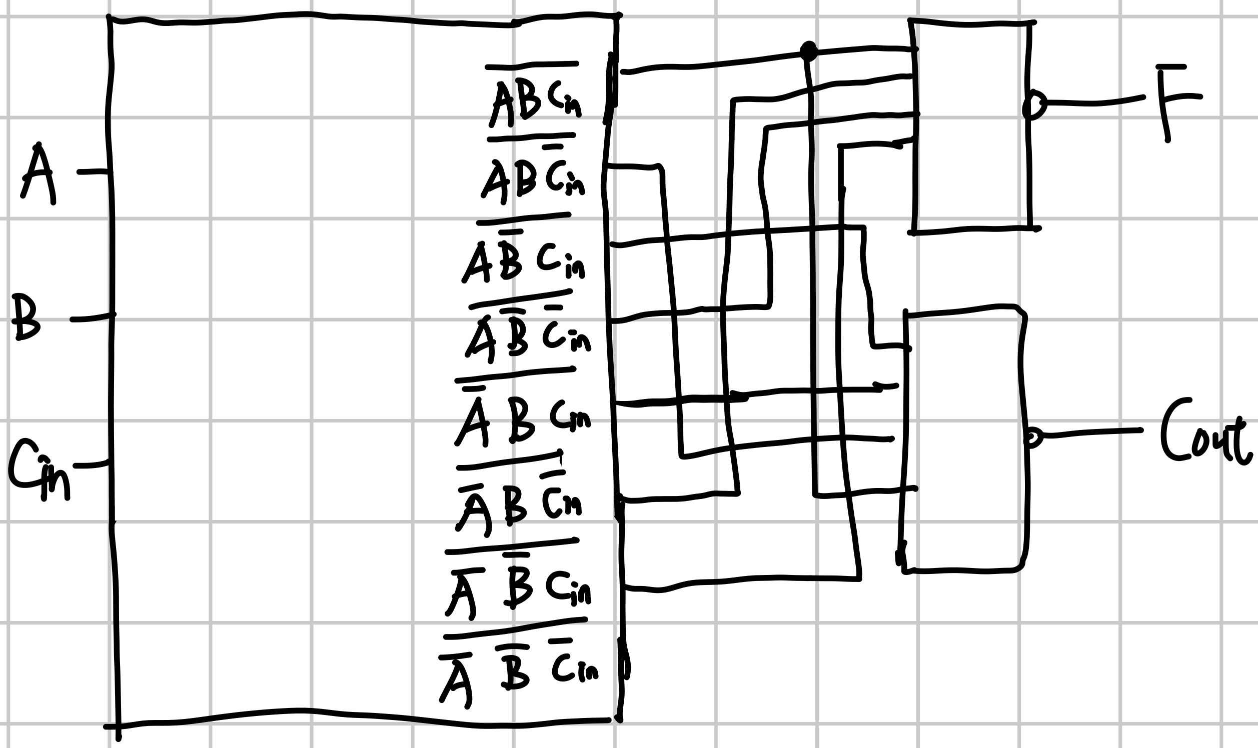
输出为 $F_0$ $C_{out}$

功能表为

| A | B | $C_{in}$ | $F_0$ | $C_{out}$ |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 |

$$F_0 = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in}$$
$$= \overline{\overline{\bar{A}\bar{B}C_{in}}\ \overline{\bar{A}B\bar{C}_{in}}\ \overline{A\bar{B}\bar{C}_{in}}\ \overline{ABC_{in}}}$$

$$C_{out} = \bar{A}BC_{in} + A\bar{B}C_{in} + AB\bar{C}_{in} + ABC_{in}$$
$$= \overline{\overline{\bar{A}BC_{in}}\ \overline{A\bar{B}C_{in}}\ \overline{AB\bar{C}_{in}}\ \overline{ABC_{in}}}$$



一位全减器 . A-B.

功能表为

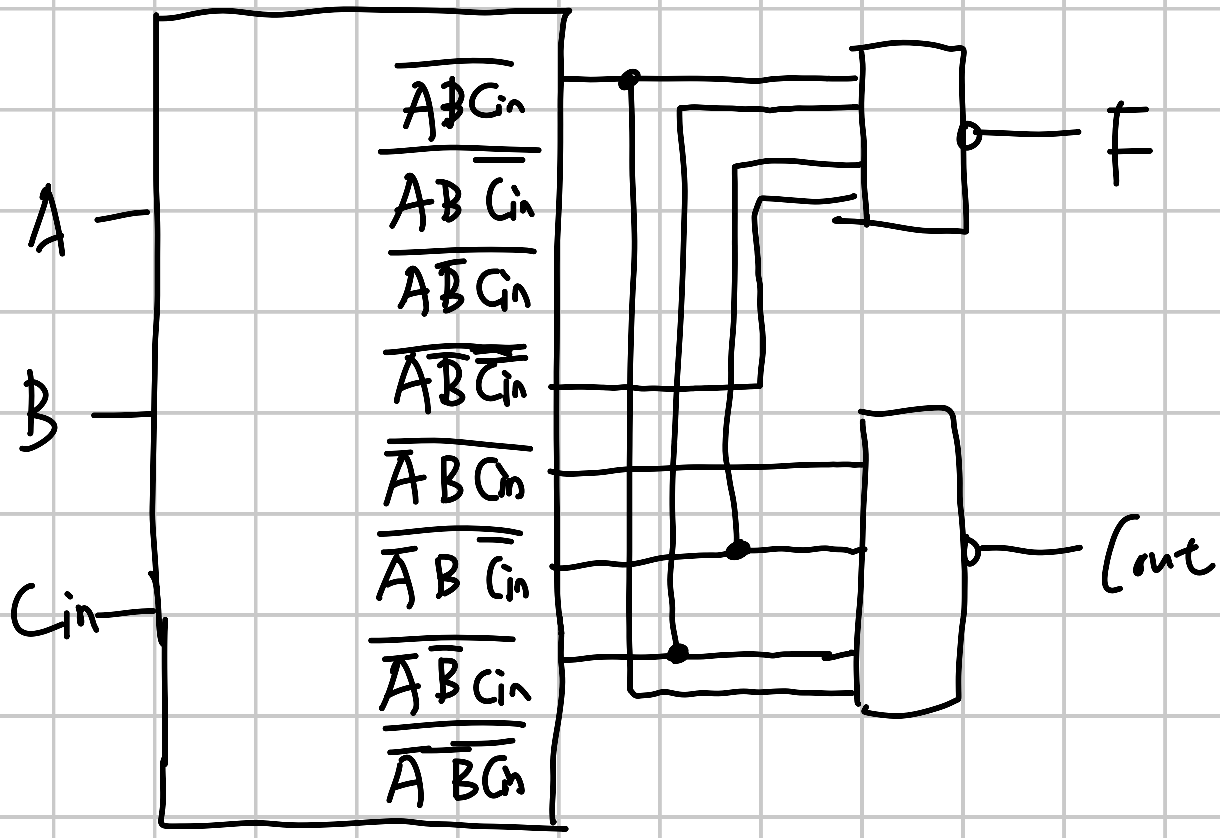
| A | B | $C_{in}$ | F | $C_{out}$ |
| --- | --- | --- | --- | --- |
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 1 |
| 0 | 1 | 0 | 1 | 1 |
| 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 |

$$F=\bar{A}\bar{B}C_{in}+\bar{A}B\bar{C}_{in}+A\bar{B}\bar{C}_{in}+ABC_{in}$$

$$=\overline{\overline{\bar{A}\bar{B}C_{in}}\;\overline{\bar{A}B\bar{C}_{in}}\;\overline{A\bar{B}C_{in}}\;\overline{ABC_{in}}}$$

$$C_{out}=\overline{AB}C_{in}+\bar{A}B\bar{C}_{in}+\bar{A}BC_{in}+ABC_{in}$$

$$=\overline{\overline{\overline{AB}C_{in}}\;\overline{\bar{A}B\bar{C}_{in}}\;\overline{\bar{A}BC_{in}}\;\overline{ABC_{in}}}$$

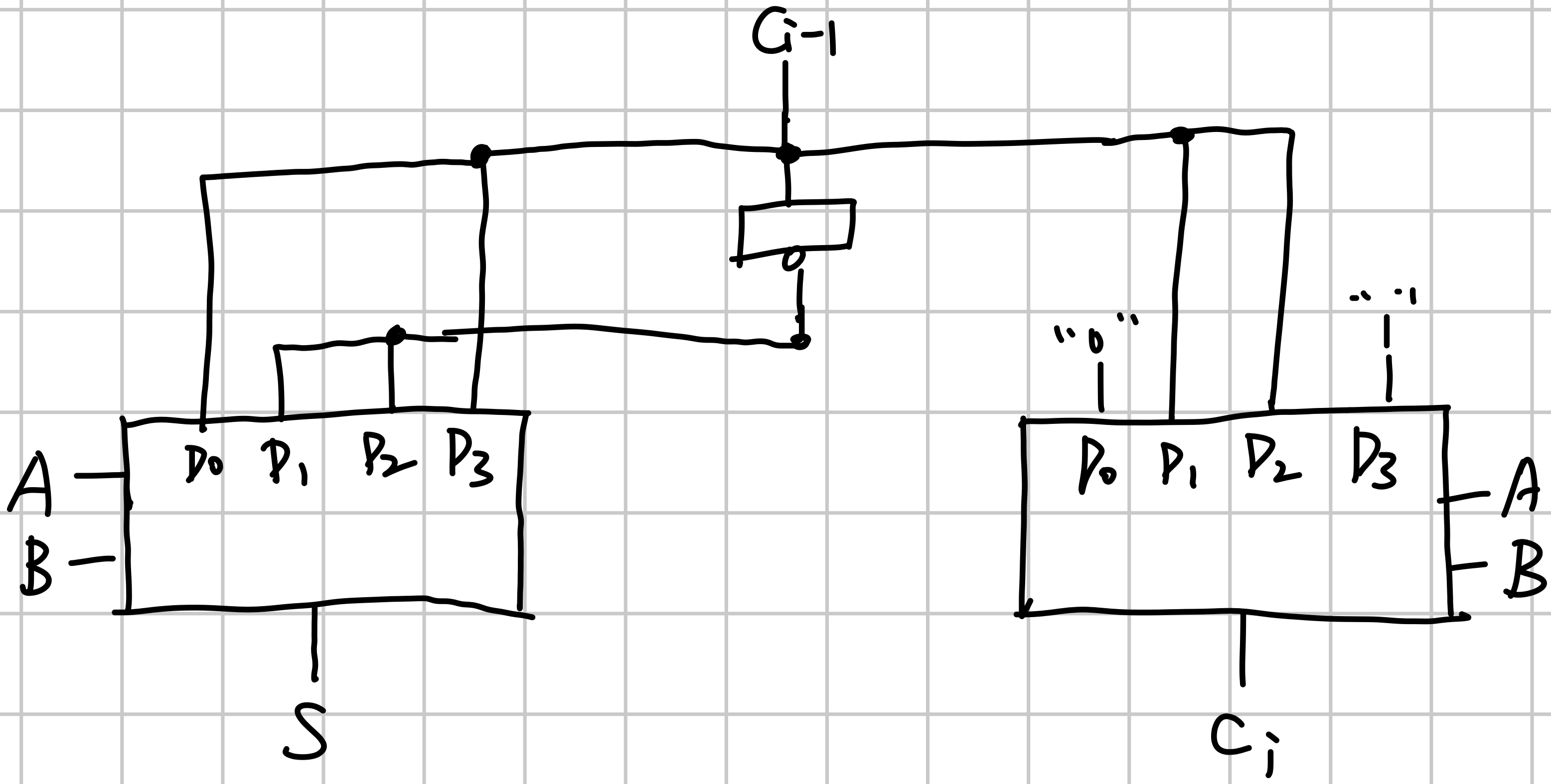


4.23

由4.18可知，

$$S = ABC_{i-1} + A\overline{B}\,\overline{C_{i-1}} + \overline{A}B\overline{C_{i-1}} + \overline{AB}C_{i-1}$$

$$C_i = \overline{A}BC_{i-1} + A\overline{B}C_{i-1} + AB\overline{C_{i-1}} + ABC_{i-1}$$



# 附加题

① 1位全加器构成16位串行加法器.

用(c)型和(d)型一级全加器交叉串联形成16位全加器应该是同类型中最快的

即(c)的 $\overline{C_n}$ 输出到d中, (d)的 $C_n$ 输出到(c)中.

已知(c)(d)型一位全加器单独获得c、F的门延时分别为2级、3级

(c)型全加器获得 $\overline{C}$ 需一级门延时, 可知

$C_1=2 \quad F_1=3$

$F_i=C_i+1$

$C_{2i}=\overline{C_{2i-1}}+1=C_{2i-1}$

$C_{2i+1}=C_{2i}+2$

得 $C_i=\lfloor\frac{i+1}{2}\rfloor\times 2$

$C_{16}=16$

$F_{16}=C_{16}+1=17$

∴ 1位全加器串联形成的最快的16位串行加法器的门延时为 $C_{16}=6$ $F_{16}=17$

② 4位并行全加器构成16位串行全加器

4位并行全加器 $C_i$ 延迟级数与位数无关, 都是2级.

$F_i$ 基本都是3级

用4片四位并行全加器组成16位加法器,

片间进位串行逐片传递

∴ $C_1\sim C_4=2 \quad F_1\sim F_4=3$

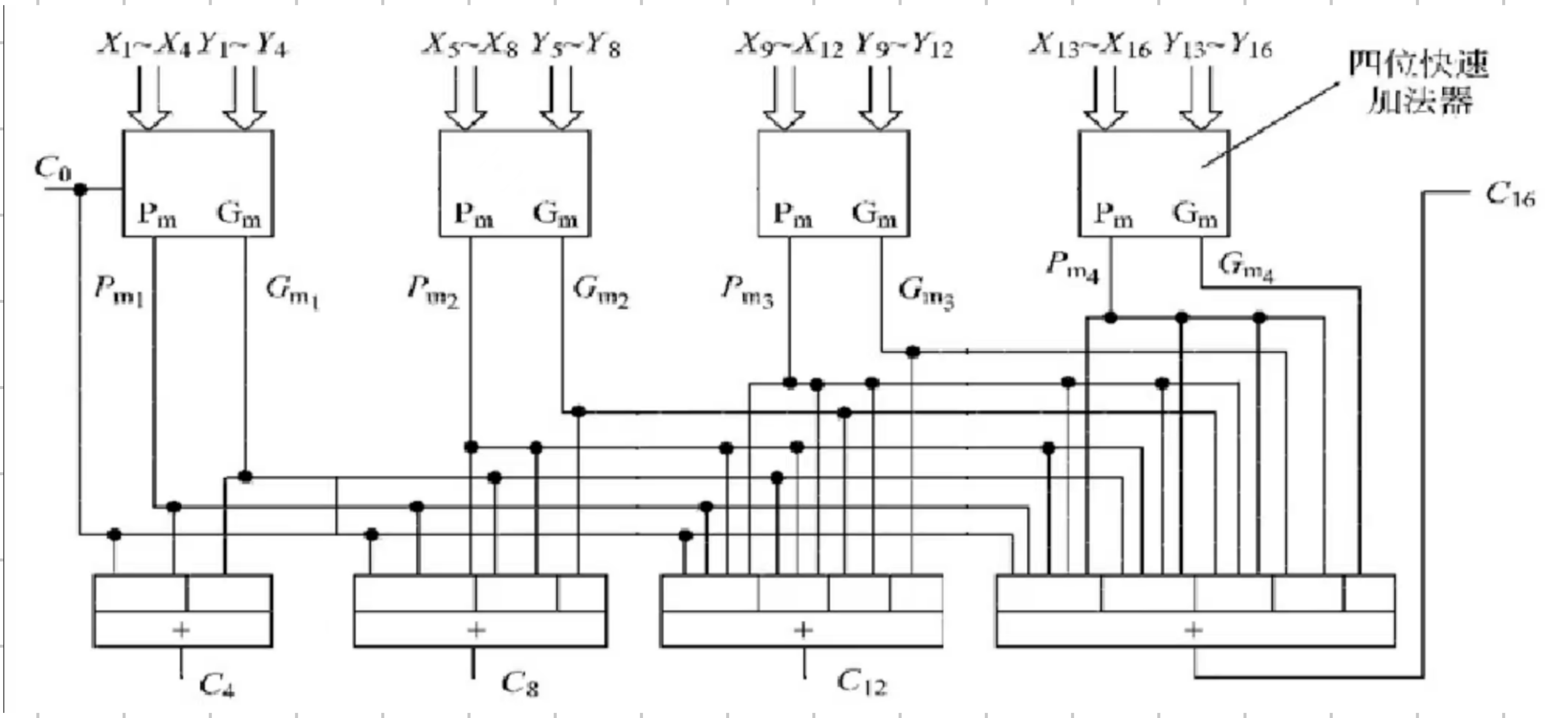
$C_5=C_4+2=4 \quad F_5=C_5+1=5 \quad C_5\sim C_8=4 \quad F_5\sim F_8=5$

$C_9=C_8+2=6 \quad F_9=C_9+1=7 \quad C_9\sim C_{12}=6 \quad F_9\sim F_{12}=7$

$C_{13}=C_{12}+2=8 \quad F_{13}=C_{13}+1=9 \quad C_{13}\sim C_{16}=8 \quad F_{13}\sim F_{16}=9$

最终结果 $F_{16}=9 \quad C_{16}=8$

③ 16位并行加法器




四位并行加法器的输出提供 $P_m$. $G_m$. 需2级延迟. F需3级延迟

$\therefore C_4 = C_8 = C_{12} = C_{16} = 2+1+3 = 3$ 级延迟

$F_1 \sim F_4 = 3$ $F_5 \sim F_8 = C_4 + 3 = 6$

$F_9 \sim F_{12} = C_8 + 3 = 6$ $F_{13} \sim F_{16} = C_{12} + 3 = 6$

$\therefore C_{16} = 3$ $F_{16} = 6$